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EX NO: 9 SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF ENCODED OUTPUT BIT SEQUENCES USING MATLAB

Objectives

(i) To implement convolutional encoding for a given binary input sequence using a rate convolutional encoder and tabulate the corresponding encoded output pairs.

(ii) To visualize and compare the input bit sequence and encoded output bit sequence using MATLAB plots to analyze the redundancy introduced by convolutional coding.

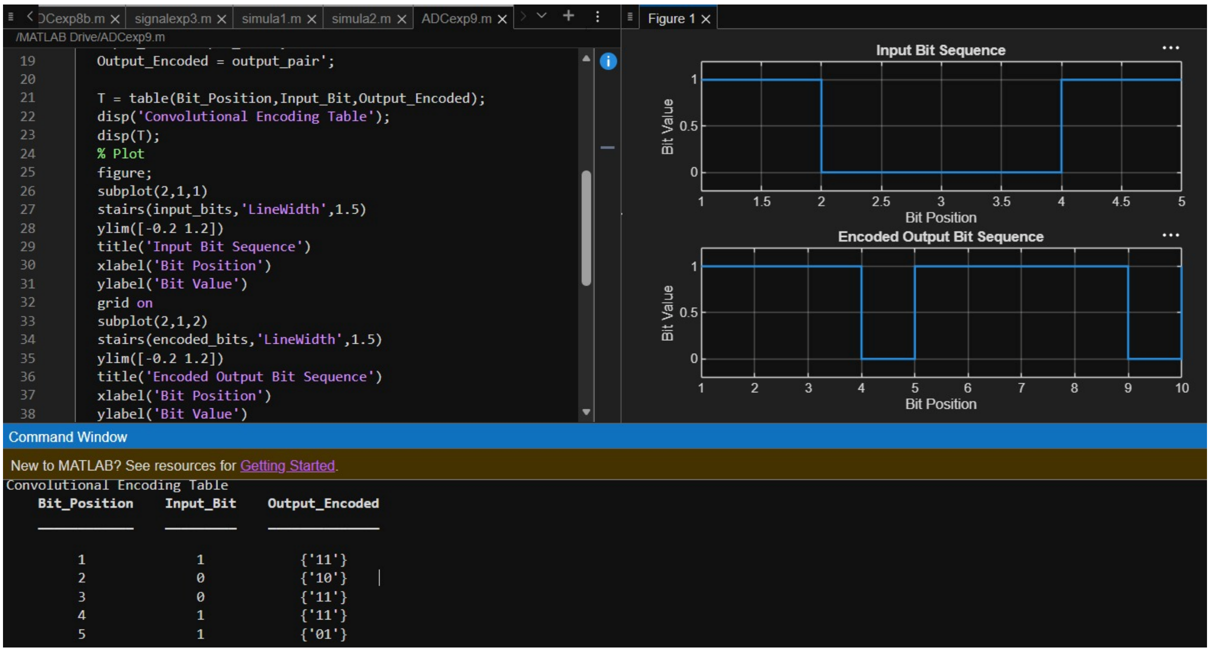
MATLAB code

```
clc;
clear;
close all;
% Fixed input sequence
input_bits = [1 0 0 1 1];
% Convolutional encoder
trellis = poly2trellis(3,[7 5]);
encoded_bits = convenc(input_bits,trellis);
% Separate encoded bits into pairs
encoded_pairs = reshape(encoded_bits,2,[])';
% Create output strings
for i = 1:length(input_bits)
output_pair{i} = sprintf('%d%d', ...
encoded_pairs(i,1),encoded_pairs(i,2));
end
% Create table
```

```
Bit_Position = (1:length(input_bits))';
Input_Bit = input_bits';
Output_Encoded = output_pair';

T = table(Bit_Position,Input_Bit,Output_Encoded);
disp('Convolutional Encoding Table');
disp(T);
% Plot
figure;
subplot(2,1,1)
stairs(input_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Input Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on
subplot(2,1,2)
stairs(encoded_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Encoded Output Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on
```

OUTPUT



Result:

The input and encoded bit sequences were plotted using MATLAB. The encoded sequence contained 10 bits for 5 input bits, confirming a rate convolutional code. The plots clearly demonstrated the redundancy introduced by convolutional encoding, which enhances the error- correction capability of digital communication systems.